

Operating Systems — Assignment 1			
Name: _____			
Roll Number: _____			
Section: _____			
Requirements	Description	Total Marks	Marks Obtained
Question 1: Cleanup Script			
Identify largest files	List the 10 largest files and prompt for deletion.	7	
Detect duplicate files	Find duplicates across directories.	7	
Delete/Hard-link/Move duplicates	Provide options for handling duplicates.	4	
Clean system junk files	Clear cache, logs, and unused packages.	7	
Bonus: Access time suggestion	Suggest deletable files based on access time.	5	
	Total Marks for Q1:	25	
Question 2: Menu-based File Operations			
Option 1: File permissions	Show and invert file permissions.	5	
Option 2: Search with wildcards	Search file for a string (supporting dot wildcard).	6	
Option 3: Content aggregation	Merge file contents (even/odd line logic).	7	
Option 4: File modification date	Check and update modified date with log maintenance.	7	
	Total Marks for Q2:	25	
Question 3: Application Tracker			
Monitor active applications	Track the active application every minute.	7	
Log usage time	Log time spent on each app.	7	
Generate usage reports	Create daily/weekly usage reports.	7	
Alert on non-productive apps	Warn if an app is used excessively.	4	
Bonus: App blocking	Block apps after exceeding a time limit.	5	
	Total Marks for Q3:	25	
Question 4: String Operations with Process Tree			
Create process tree	Fork child processes for each operation.	7	
Use of Exec syscall	Exec syscall for each of 5 operations	6	
Tasks performed by Child Processes	Implement 5 tasks in separate file for each child	5	
Log task details & results	Save task info and results to a file.	2	
	Total Marks for Q4:	20	
Question 5: Expense Tracker			
Add expenses & income	Input expenses, category, amount, and date.	4	
Categorize transactions	Auto-classify transactions (food, bills, etc.).	4	
Calculate summaries	Show category totals, savings, and expenses.	3	
Budget alert	Warn when spending crosses a budget limit.	2	
Export to CSV	Save transaction data to CSV format.	2	
Bonus: ASCII graphs	Visualize spending trends (using tools like gnuplot).	5	
	Total Marks for Q5:	15	
	Overall Total:	110	